CBSE Class 12 Physics — Half-Yearly Predicted Paper (Set 7 of 10)

Section A: 1-Mark Questions (MCQ/Assertion-Reason)

Q1. In an experiment, three microscopic latex spheres are sprayed into a chamber and become charged with charges $+3e$, $+5e$, and $-3e$ respectively. All three spheres came in contact simultaneously for a moment and got separated. Which one of the following are possible values for the final charge on the spheres? (a) $+5e, -1e, +5e$ (b) $+6e, +6e, -7e$ (c) $-4e, +3.5e, +5.5e$ (d) $+5e, -8e, +7e$ [PYQ 2021] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q2. Two point charges $+8q$ and $-2q$ are located at $x = 0$ and $x = L$ respectively. The point on the x-axis at which the net electric field is strictly zero due to these charges is: (a) $8L$ (b) $4L$ (c) $2L$ (d) L [PYQ 2021] | **[Chapter 1: Electric Charges & Fields | Confidence: High]**

Q3. Two parallel plate capacitors X and Y have the exact same area of plates and same separation between plates. X has air and Y contains a dielectric of constant $K = 2$ between its plates. They are connected in series to a battery of $12\ \mathrm{V}$. The ratio of electrostatic energy stored in X and Y ($U_X : U_Y$) is: (a) $4:1$ (b) $1:4$ (c) $2:1$ (d) $1:2$ [PYQ 2021] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q4. An electric current is passed through a circuit containing two wires of the same material, connected in parallel. If the lengths and radii of the wires are in the ratio of $3:2$ and $2:3$ respectively, then the ratio of the current passing through the wires ($I_1 : I_2$) will be: (a) $2:3$ (b) $3:2$ (c) $8:27$ (d) $27:8$ [PYQ 2021] | **[Chapter 3: Current Electricity | Confidence: High]**

Q5. A conductor of $10\ \Omega$ is connected across a $6\ \mathrm{V}$ ideal voltage source. The electrical power supplied by the source to the conductor is: (a) $1.8\ \mathrm{W}$ (b) $2.4\ \mathrm{W}$ (c) $3.6\ \mathrm{W}$ (d) $7.2\ \mathrm{W}$ [PYQ 2023] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q6. An electron with an orbital angular momentum L moving around the nucleus has a magnetic dipole moment given by: (a) $eL/2m$ (b) $eL/3m$ (c) $eL/4m$ (d) eL/m [PYQ 2023] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q7. A long straight solid wire of circular cross-section of radius a carries a steady current I. The current is uniformly distributed across its cross-section. The ratio of the magnitudes of the magnetic field at a point distant $a/2$ from the axis (inside) to that at a point distant $2a$ from the axis (outside) is: (a) $4:1$ (b) $1:1$ (c) $4:3$ (d) $3:4$ [PYQ 2022] | **[Chapter 4: Moving Charges & Magnetism | Confidence: High]**

Q8. The magnetic susceptibility (χ) of a given magnetic material is -4.2×10^{-6}. The nature of the material is: (a) Ferromagnetic (b) Paramagnetic (c) Diamagnetic (d) Non-magnetic [PYQ 2022] | **[Chapter 5: Magnetism & Matter | Confidence: Very High]**

Q9. A conducting square loop of side L and resistance R moves in its plane with a uniform velocity v perpendicular to one of its sides. A magnetic induction B, constant in time and space, pointing perpendicular and into the plane of the loop exists everywhere. The induced current in the loop is: (a) BLv/R Clockwise (b) BLv/R Anticlockwise (c) $2BLv/R$ Anticlockwise (d) Zero [PYQ 2021] | **[Chapter 6: Electromagnetic Induction | Confidence: High]**

Q10. The instantaneous values of emf and current in a series AC circuit are $E = E_0 \sin(\omega t)$ and $I = I_0 \sin(\omega t + \pi/3)$ respectively. The circuit is: (a) necessarily an RL circuit. (b) necessarily an RC circuit. (c) necessarily an LCR circuit. (d) can be an RC or LCR circuit. [PYQ 2021] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q11. Electromagnetic waves are often used to improve visibility on airport runways during heavy fog and mist conditions. Which region of the electromagnetic spectrum do these radiations belong to? (a) Ultraviolet rays (b) Infrared rays (c) Microwaves (d) X-rays [PYQ 2022] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q12. The kinetic energy of a proton and that of an α-particle are $4\ \mathrm{eV}$ and $1\ \mathrm{eV}$, respectively. The ratio of the de-Broglie wavelengths associated with them ($\lambda_p : \lambda_\alpha$) will be: (a) $2:1$ (b) $1:1$ (c) $1:2$ (d) $4:1$ [PYQ 2020] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

For Questions 13 to 16, two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c), and (d) as given below. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) Both A and R are false.

Q13. Assertion (A): A negative charge in an electric field naturally moves exactly along the direction of the electric field. **Reason (R):** On a negative charge, an electrostatic force acts perfectly in the direction of the applied electric field. [PYQ 2021] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q14. Assertion (A): A wire bent into an irregular shape with points P and Q fixed. If a steady current I is passed through the wire, the area enclosed by the irregular portion of the wire tends to increase. **Reason (R):** Opposite currents carrying wires repel each other. [PYQ 2022] | **[Chapter 4: Moving Charges & Magnetism | Confidence: High]**

Q15. Assertion (A): In an interference pattern observed in Young's double slit experiment, if the separation (d) between coherent sources as well as the distance (D) of the screen from the sources both are reduced to $1/3^{\mathrm{rd}}$, then the new fringe width remains exactly the same. **Reason (R):** The fringe width in Young's double slit experiment is directly proportional to the ratio (d/D). [PYQ 2022] | **[Chapter 10: Wave Optics | Confidence: High]**

Q16. Assertion (A): The photoelectrons produced by a monochromatic light beam incident on a metal surface have a wide spread in their kinetic energies, ranging from zero to a maximum value. **Reason (R):** The energy of electrons emitted from deeper inside the metal surface is lost in collisions with the other atoms in the metal lattice before they can escape. [PYQ 2022] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Section B: 2-Mark Questions (Very Short Answer)

Q17. Two identical cells of emfs $1.5\ \mathrm{V}$ and $2.0\ \mathrm{V}$ having internal resistances $0.2\ \Omega$ and $0.3\ \Omega$ respectively, are connected in parallel with their positive terminals joined together. Calculate the exact equivalent emf of this parallel combination. [PYQ 2016] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q18. Two identical circular loops P and Q of radius $5\ \mathrm{cm}$ each are lying in mutually perpendicular planes with a common centre. Find the magnitude of the net magnetic field at their common centre if they carry steady currents of $3\ \mathrm{A}$ and $4\ \mathrm{A}$ respectively. [PYQ 2023] | **[Chapter 4: Moving Charges & Magnetism | Confidence: High]**

Q19. A small bar magnet of magnetic dipole moment $6\ \mathrm{J/T}$ is aligned at an angle of 60° with a uniform external magnetic field of $0.44\ \mathrm{T}$. Calculate the work done in turning the magnet to align its magnetic moment normal (90°) to the applied magnetic field. [PYQ 2023] | **[Chapter 5: Magnetism & Matter | Confidence: Very High]**

Q20. A flexible conducting wire of irregular shape, as shown in standard diagrams, forms into a perfect circular shape when placed in a region of a strong magnetic field which is directed normal to the plane of the loop pointing strictly away from the reader. Predict the direction of the induced current in the wire, justifying with Lenz's law. [PYQ 2014] | **[Chapter 6: Electromagnetic Induction | Confidence: High]**

Q21. You are provided with the following three lenses: L_1: Power = $3\ \mathrm{D}$, Aperture = $8\ \mathrm{cm}$ L_2: Power = $6\ \mathrm{D}$, Aperture = $1\ \mathrm{cm}$ L_3: Power = $16\ \mathrm{D}$, Aperture = $1\ \mathrm{cm}$ Which two lenses will you specifically select to use as an eyepiece and as an objective to construct an astronomical refracting telescope? Give a clear optical reason. [PYQ 2017] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Section C: 3-Mark Questions (Short Answer)

Q22. Two small identical conducting spheres carrying charges of $+10\ \mu\mathrm{C}$ and $-20\ \mu\mathrm{C}$ are separated by a distance r in air, experiencing an attractive electrostatic force F. If they are brought into direct physical contact and then separated to a new distance of $r/2$, what will be the exact nature and magnitude of the new force between them in terms of the initial force F? [PYQ 2021] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q23. Define the term 'conductivity' of a metallic wire. Using the physical concept of free electrons in a conductor, derive the mathematical expression for the electrical conductivity (σ) of a wire in terms of its number density (n), elementary charge (e), relaxation time (τ), and mass of the electron (m). [PYQ 2018] | **[Chapter 3: Current Electricity | Confidence: High]**

Q24. Two long, straight, parallel conductors separated by a distance d carry steady currents I_1 and I_2 in the exact same direction. (A) Draw the magnetic field lines representing the field produced by these conductors. (B) Derive the mathematical expression for the magnetic force per unit length acting between them. Is this force attractive or repulsive? [PYQ 2016, 2022] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q25. A series LCR circuit consisting of a $2\ \mu\mathrm{F}$ capacitor, a $100\ \Omega$ resistor, and an $8\ \mathrm{H}$ inductor is connected in series with an AC source. (A) What should be the source frequency such that the current drawn in the circuit is maximum? (B) What is this specific condition called? (C) If the peak value of the applied emf is $200\ \mathrm{V}$, find the maximum current flowing in the circuit at this frequency. [PYQ 2016] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q26. Identify the specific electromagnetic wave whose wavelength varies in the respective ranges: (A) $10^{-12}\ \mathrm{m} < \lambda < 10^{-8}\ \mathrm{m}$ (B) $10^{-3}\ \mathrm{m} < \lambda < 10^{-1}\ \mathrm{m}$ Write one critical practical application for each of these two electromagnetic waves. [PYQ 2017] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q27. A slit of width $a = 0.6\ \mathrm{mm}$ is illuminated by a beam of light consisting of two closely spaced wavelengths $600\ \mathrm{nm}$ and $480\ \mathrm{nm}$. The resulting diffraction pattern is observed on a screen placed exactly $1.0\ \mathrm{m}$ from the slit. Find the distance of the second secondary bright fringe from the central maximum on the screen, specifically pertaining to the light of wavelength $600\ \mathrm{nm}$. [PYQ 2022] | **[Chapter 10: Wave Optics | Confidence: High]**

Q28. State Einstein's photoelectric equation explaining all the physical symbols used. A graph of the maximum kinetic energy (K_{max}) of the emitted photoelectrons versus the frequency (ν) of incident light is a straight line. Explain mathematically how you can determine the value of Planck's constant (h) and the work function (Φ) of the photosensitive material directly from this graph. [PYQ 2018] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Section D: 4-Mark Questions (Case-Study Based)

Q29. Case Study: The Wheatstone Bridge A Wheatstone bridge is an electrical circuit network used to accurately measure an unknown electrical resistance by balancing two legs of a bridge circuit. The bridge consists of four resistors $P, Q, R,$ and S connected to form a quadrilateral. A galvanometer is connected between two opposite junctions, and a battery is connected between the other two. When the potential difference across the galvanometer becomes zero, no current flows through it, and the bridge is said to be completely balanced. This highly sensitive null-point method provides extremely accurate measurements unaffected by the internal resistance of the driving battery. (A) Write the standard mathematical condition for a Wheatstone bridge to be perfectly balanced. (1 Mark) (B) Why is it generally advised to construct the resistive wires of such practical measuring bridges (like the Meter Bridge) using alloys such as manganin or constantan instead of pure copper? (1 Mark) (C) In a particular Wheatstone bridge arrangement, the four arms have resistances $P = 10\ \Omega$, $Q = 10\ \Omega$, $R = 10\ \Omega$, and $S = 20\ \Omega$. Calculate the exact value of the shunt resistance that must be connected strictly in parallel to the arm S to balance the bridge. (2 Marks) [Practice / PYQ 2022] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q30. Case Study: Refraction Through a Prism When a monochromatic ray of light passes through a glass prism, it suffers refraction at two distinct inclined planar faces, deviating the ray from its original path. The total angle of deviation (δ) depends heavily on the angle of incidence (i), the angle of the prism (A), and the refractive index (μ) of the prism material. As the angle of incidence is steadily increased from a very small value, the angle of deviation initially decreases, reaches a strict minimum value (δ_m), and then starts increasing again. (A) What is the specific geometric condition inside the prism (relating angles r_1 and r_2) for the ray to suffer minimum deviation? (1 Mark) (B) How does the minimum angle of deviation (δ_m) conceptually change if the incident violet light is completely replaced by red light? (1 Mark) (C) A monochromatic ray of light incident at an angle of 45° on the face of an equilateral glass prism (Angle of prism $A = 60^\circ$) suffers minimum deviation. Calculate the refractive index (μ) of the material of the prism. (2 Marks) [PYQ 2017, 2020] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Section E: 5-Mark Questions (Long Answer)

Q31. (A) State Gauss's law in electrostatics. (B) Use Gauss's law to systematically derive the mathematical expression for the electric field intensity (E) due to an infinitely large, thin plane sheet of charge possessing a uniform surface charge density $+\sigma$. (C) Two parallel, infinitely large thin metal sheets have equal surface charge densities of exactly $26.4 \times 10^{-12}\ \mathrm{C/m^2}$ but of opposite signs. Calculate the magnitude of the electric field: (i) strictly in the outer regions surrounding the two sheets. (ii) perfectly in the space between these two sheets. [PYQ 2020, 2021] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q32. (A) State the fundamental underlying principle of an AC generator. (B) Draw a neatly labelled schematic diagram of an AC generator showing its major components (armature, field magnets, slip rings, and brushes). (C) Using Faraday's laws of electromagnetic induction, rigorously derive the mathematical expression for the instantaneous alternating emf (ε) induced in the rotating coil. (D) A circular coil consisting of 20 turns and a radius of $10\ \mathrm{cm}$ is rotated about its vertical diameter with a constant angular speed of $50\ \mathrm{rad/s}$ in a uniform horizontal magnetic field of $3.0 \times 10^{-2}\ \mathrm{T}$. Calculate the maximum (peak) emf induced in the coil. [PYQ 2017, 2019] | **[Chapter 6: Electromagnetic Induction | Confidence: Very High]**

Q33. (A) Describe Thomas Young's Double Slit Experiment (YDSE). (B) Two monochromatic light waves having displacements $y_1 = a\cos(\omega t)$ and $y_2 = a\cos(\omega t + \phi)$ from two coherent sources superimpose to produce an interference pattern on a screen. Derive the expression for the resultant intensity (I) at any point on the screen. (C) Based on this derived intensity, clearly state the phase conditions required for constructive and destructive interference. (D) In a specific YDSE setup, the ratio of the lowest intensity at the minima to the highest intensity at the maxima is found to be exactly $9 : 25$. Calculate the exact ratio of the physical widths of the two respective slits. [PYQ 2014, 2019] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Answer Key

Q1: (b) $+6e, +6e, -7e$. The total initial charge is $+3e + 5e - 3e = +5e$. By conservation of charge, the final sum must also be $+5e$. Only option (b) sums to $+5e$ ($6 + 6 - 7 = 5$). **Q2:** (c) $2L$. For opposite charges, the null point lies entirely outside the charges, closer to the smaller charge ($-2q$). Let distance be d from $-2q$ away from $8q$. Field balance: $k(8q)/(L+d)^2 = k(2q)/d^2 \Rightarrow 4/(L+d)^2 = 1/d^2 \Rightarrow 2/(L+d) = 1/d \Rightarrow L + d = 2d \Rightarrow d = L$. Position on axis = $L + L = 2L$. **Q3:** (c) $2 : 1$. In a series circuit, charge Q is identical. Energy stored $U = Q^2/2C \implies U \propto 1/C$. Capacitance $C_Y = K \cdot C_X = 2C_X$. Therefore, $U_X/U_Y = C_Y/C_X = 2C_X/C_X = 2/1$. **Q4:** (c) $8 : 27$. Resistance $R = \rho L/(\pi r^2) \implies R \propto L/r^2$. In parallel, Voltage V is same, so current $I \propto 1/R \propto r^2/L$. Ratio $I_1/I_2 = (r_1/r_2)^2 \times (L_2/L_1) = (2/3)^2 \times (2/3) = (4/9) \times (2/3) = 8/27$. **Q5:** (c) $3.6\ \mathrm{W}$. Power $P = V^2/R = (6)^2/10 = 36/10 = 3.6\ \mathrm{W}$. **Q6:** (a) $eL/2m$. The gyromagnetic ratio of an electron is strictly constant. $\mu_l/L = e/2m \implies \mu_l = eL/2m$. **Q7:** (b) $1 : 1$. Inside wire ($r < a$): $B \propto r$. At $a/2$, $B_{in} = \frac{\mu_0 I(a/2)}{2\pi a^2} = \frac{\mu_0 I}{4\pi a}$. Outside wire ($r > a$): $B \propto 1/r$. At $2a$, $B_{out} = \frac{\mu_0 I}{2\pi(2a)} = \frac{\mu_0 I}{4\pi a}$. Ratio is exactly $1 : 1$. **Q8:** (c) Diamagnetic. Diamagnetic materials strictly possess a small, negative magnetic susceptibility. **Q9:** (d) Zero. Since the magnetic field is uniform in space and completely constant in time, the total magnetic flux linked with the fully immersed moving loop does not change ($d\Phi/dt = 0$). Hence, induced emf and current are zero. **Q10:** (d) can be an RC or LCR circuit. The current leads the voltage by a phase angle of $\pi/3$ (60°). Leading current indicates a net capacitive circuit, which occurs in a pure RC circuit or an LCR circuit where $X_C > X_L$. **Q11:** (b) Infrared rays. They suffer minimal scattering due to atmospheric particles, effectively piercing through fog and mist. **Q12:** (b) $1 : 1$. $\lambda = h/\sqrt{2mK}$. Ratio $\lambda_p/\lambda_\alpha = \sqrt{(m_\alpha K_\alpha)/(m_p K_p)} = \sqrt{(4m_p \times 1\ \mathrm{eV})/(m_p \times 4\ \mathrm{eV})} = \sqrt{4/4} = 1 : 1$.

Q13: (d) Both A and R are false. A negative charge experiences force strictly in the *opposite* direction to the electric field. **Q14:** (a) Both A and R are true and R is the correct explanation of A. An irregular loop carrying current can be seen as small opposite current segments facing each other across the loop. Their mutual repulsion pushes the loop outwards, maximizing the area into a circle. **Q15:** (c) A is true but R is false. $\beta = \lambda D/d$. If $D \to D/3$ and $d \to d/3$, $\beta' = \lambda(D/3)/(d/3) = \lambda D/d = \beta$. The reason statement claims $\beta \propto d/D$, which is inverted and thus completely false. **Q16:** (a) Both A and R are true and R is the correct explanation of A.

Q17: Equivalent emf $E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} = \frac{(1.5\times0.3)+(2.0\times0.2)}{0.2+0.3} = \frac{0.45+0.40}{0.5} = \frac{0.85}{0.5} = 1.7\ \mathrm{V}$. **Q18:** Fields are perpendicular: $\vec{B}_{net} = \sqrt{B_1^2 + B_2^2}$. $B = \frac{\mu_0 I}{2R}$. $B_{net} = \frac{\mu_0}{2R}\sqrt{3^2+4^2} = \frac{\mu_0}{2(0.05)}\sqrt{25} = \frac{5\mu_0}{0.1} = 50\mu_0\ \mathrm{T}$ (or $50 \times 4\pi \times 10^{-7} = 2\pi \times 10^{-5}\ \mathrm{T}$). **Q19:** Work done $W = MB(\cos\theta_1 - \cos\theta_2) = 6 \times 0.44 \times (\cos 60^\circ - \cos 90^\circ) = 2.64 \times (0.5 - 0) = 1.32\ \mathrm{Joules}$. **Q20:** Expanding into a circle critically increases the area and thus increases the inward magnetic flux. According to Lenz's law, the induced current must strictly oppose this increase by generating an outward magnetic field. To produce an outward field, the current must explicitly flow in the **anticlockwise** direction. **Q21:** Objective: L_1 (Lowest power $3\ \mathrm{D} \implies$ highest focal length for magnification, and largest aperture $8\ \mathrm{cm}$ for maximum light gathering power). Eyepiece: L_3 (Highest power $16\ \mathrm{D} \implies$ shortest focal length to yield maximum magnification $M = f_o/f_e$).

Q22: Initial force $F = \frac{k(10\mu\mathrm{C})(-20\mu\mathrm{C})}{r^2} = -\frac{200k\mu^2}{r^2}$ (Attractive). Upon contact, net charge $= 10 - 20 = -10\ \mu\mathrm{C}$. They share it equally, so each gets $-5\ \mu\mathrm{C}$. New force $F' = \frac{k(-5\mu\mathrm{C})(-5\mu\mathrm{C})}{(r/2)^2} = \frac{25k\mu^2}{r^2/4} = \frac{100k\mu^2}{r^2}$ (Repulsive). Ratio $F'/|F| = 100/200 = 1/2$. New force is **Repulsive** and magnitude is exactly $F/2$. **Q23:** Current $I = neAv_d$. Drift velocity $v_d = \frac{eE\tau}{m}$. Substituting, $I = neA(\frac{eE\tau}{m}) = \frac{ne^2A\tau}{m}E$. Current density $J = I/A = \frac{ne^2\tau}{m}E$. By Ohm's law, $J = \sigma E$. Therefore, conductivity $\sigma = \frac{ne^2\tau}{m}$. **Q24:** (A) Draw parallel straight wires. Magnetic field lines are concentric circles around each wire. (B) Field due to wire 1 at wire 2 is $B_1 = \frac{\mu_0 I_1}{2\pi d}$. Force on length L of wire 2 is $F = I_2 L B_1 \sin(90^\circ) = \frac{\mu_0 I_1 I_2 L}{2\pi d}$. Force per unit length $f = F/L = \frac{\mu_0 I_1 I_2}{2\pi d}$. By Fleming's Left-Hand Rule, the force is definitively **Attractive**. **Q25:** (A) Current is strictly maximum at electrical resonance, $\omega_r = 1/\sqrt{LC} \implies f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{8\times2\times10^{-6}}} = \frac{1}{2\pi\times4\times10^{-3}} = \frac{1000}{8\pi} \approx 39.8\ \mathrm{Hz}$. (B) Electrical Resonance. (C) At resonance, impedance $Z = R = 100\ \Omega$. Max peak current $I_0 = V_0/Z = 200/100 = 2.0\ \mathrm{A}$. **Q26:** (A) $10^{-12}\ \mathrm{m} < \lambda < 10^{-8}\ \mathrm{m}$ specifies **X-rays**. Application: Used extensively as a diagnostic tool in medicine (radiography) to examine bone fractures. (B) $10^{-3}\ \mathrm{m} < \lambda < 10^{-1}\ \mathrm{m}$ specifies **Microwaves**. Application: Used in radar systems for aircraft navigation and microwave ovens. **Q27:** For a single slit diffraction, the position of the n^{th} secondary maximum is $y_n = (n + \frac{1}{2})\frac{\lambda D}{a}$. For the second secondary bright fringe ($n = 2$), $y_2 = \frac{5\lambda D}{2a}$. $y_2 = \frac{5\times(600\times10^{-9}\ \mathrm{m})\times1.0\ \mathrm{m}}{2\times(0.6\times10^{-3}\ \mathrm{m})} = \frac{3000\times10^{-9}}{1.2\times10^{-3}} = 2500 \times 10^{-6}\ \mathrm{m} = 2.5\ \mathrm{mm}$. **Q28:** Einstein's photoelectric equation: $K_{max} = h\nu - \Phi$ (where K_{max} is max kinetic energy, h is Planck's constant, ν is incident frequency, Φ is the work function). This resembles the straight-line equation $y = mx + c$. By plotting K_{max} on the y-axis and ν on the x-axis, the specific **slope** of the resulting straight line directly yields Planck's constant (h), and the negative magnitude of the **y-intercept** precisely gives the material's work function (Φ).

Q29: (A) $P/Q = R/S$. (B) Alloys like constantan have an extremely low temperature coefficient of resistivity (α), ensuring their resistance does not dramatically drift due to Joule heating during experiments. (C) For balance, $P/Q = R/S_{eq} \implies 10/10 = 10/S_{eq} \implies S_{eq} = 10\ \Omega$. The arm has $S = 20\ \Omega$. Let shunt be x. $1/S_{eq} = 1/20 + 1/x \implies 1/10 = 1/20 + 1/x \implies 1/x = 1/10 - 1/20 = 1/20 \implies x = 20\ \Omega$. **Q30:** (A) The refracted ray inside becomes perfectly parallel to the base of the prism, resulting in symmetric conditions $r_1 = r_2 = A/2$ and $i = e$. (B) The refractive index for red light is strictly less than that for violet light ($\mu_R < \mu_V$). Since $\delta_m \propto (\mu - 1)A$ for thin prisms, red light suffers a much strictly smaller minimum deviation. (C) At minimum deviation, $r = A/2 = 60/2 = 30^\circ$. By Snell's Law, $\mu = \sin i/\sin r = \sin(45^\circ)/\sin(30^\circ) = (1/\sqrt{2})/(1/2) = 2/\sqrt{2} = \sqrt{2} \approx 1.414$.

Q31: (A) The total electric flux passing through any completely closed surface is exactly equal to $1/\epsilon_0$ times the net charge absolutely enclosed by the surface ($\oint \vec{E} \cdot d\vec{A} = Q_{enc}/\epsilon_0$). (B) Consider a pillbox (cylinder) Gaussian surface piercing the sheet. Flux exclusively escapes through the two flat circular ends of area A. $\Phi = EA + EA = 2EA$. Enclosed charge $Q = \sigma A$. By Gauss's Law: $2EA = \sigma A/\epsilon_0 \implies E = \sigma/2\epsilon_0$. (C) (i) Outside regions: The fields strictly oppose each other ($\frac{\sigma}{2\epsilon_0} - \frac{\sigma}{2\epsilon_0}$), making the net electric field identically zero. (ii) Between the sheets: The fields powerfully add up ($\frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{\epsilon_0}$). $E = \frac{26.4\times10^{-12}}{8.85\times10^{-12}} \approx 2.98\ \mathrm{V/m}$ (or $\mathrm{N/C}$). **Q32:** (A) Electromagnetic Induction. Mechanical work used to rotate a coil in a magnetic field induces an alternating electric current. (B) Draw diagram with N/S poles, rectangular armature coil, two separate slip rings (R_1, R_2), and two stationary carbon brushes (B_1, B_2). (C) Magnetic flux $\Phi = NBA\cos(\theta) = NBA\cos(\omega t)$. By Faraday's Law, $\varepsilon = -d\Phi/dt = -NBA\frac{d}{dt}\cos(\omega t) = NBA\omega\sin(\omega t)$. Thus $\varepsilon = \varepsilon_0\sin(\omega t)$. (D) Maximum peak emf $\varepsilon_0 = NBA\omega = 20 \times (3.0 \times 10^{-2}) \times (\pi \times 0.1^2) \times 50 = 20 \times 0.03 \times 0.01\pi \times 50 = 600 \times 0.0003\pi \times 50 = 18000 \times 0.0005\pi$. **Q33:** (A) Monochromatic light from a single source illuminates two narrow, closely spaced coherent slits (S_1, S_2). The expanding secondary wavelets superimpose onto a screen to form alternate equally-spaced bright and dark fringes. (B) $y = y_1 + y_2 = a\cos\omega t + a\cos(\omega t + \phi) = 2a\cos(\phi/2)\cos(\omega t + \phi/2)$. Amplitude $R = 2a\cos(\phi/2)$. Intensity $I \propto R^2 \implies I = 4I_0\cos^2(\phi/2)$. (C) Constructive: $\cos^2(\phi/2) = 1 \implies \phi = 2n\pi$. Destructive: $\cos^2(\phi/2) = 0 \implies \phi = (2n+1)\pi$. (D) $I_{min}/I_{max} = (a_1 - a_2)^2/(a_1 + a_2)^2 = 9/25$. Root yields $(a_1 - a_2)/(a_1 + a_2) = 3/5 \implies 5a_1 - 5a_2 = 3a_1 + 3a_2 \implies 2a_1 = 8a_2 \implies a_1/a_2 = 4/1$. Since intensity $I \propto a^2 \propto \mathrm{width}\ (w)$, Ratio of widths $w_1/w_2 = a_1^2/a_2^2 = (4)^2/(1)^2 = 16 : 1$.